
SUSTAINABLE ENGINEERING: **FOOD-FOCUSED PERSONAL CARBON** **FOOTPRINT CALCULATOR**

Name: Noel Joseph Kurian

Class: S3 CSD

UID: U2503216

Roll no.: 7

Submitted on: 08/09/2026

ABSTRACT

High carbon emissions are a major environmental challenge caused by activities such as transportation, consumption of Energy, food production and consumption, and waste generation. While we only talk about the carbon emissions at the industrial level, individual lifestyle choices also contribute to environmental impact. However, many people lack a simple way to understand the carbon emissions associated with their daily activities.

This report proposes a **Food-Focused Personal Carbon Footprint Calculator for College Students**. This web-based system can estimate emissions from food consumption, food waste, transportation, electricity, waste, and general consumption, with particular emphasis on food-related activities due to their relevance to **SDG 2 - Zero Hunger**. Sustainable food production and responsible consumption are important aspects of developing resilient food systems, while food waste also results in unnecessary use of resources and associated greenhouse-gas emissions.

This system follows a **Measure-Analyse-Educate-Recommend** approach. Users can enter information about their lifestyle, receive an estimated carbon footprint, identify their major sources of emissions, and receive practical recommendations for reducing their environmental impact. The system can be implemented using HTML, CSS, and JavaScript without requiring advanced technologies. This solution provides a simple, interactive, and feasible method of promoting sustainable behaviour among college students.

INTRODUCTION

Carbon emissions are produced through different everyday activities, such as transportation, electricity use, food production, consumption, and waste management. Although large-scale action by governments and industries have been taken, individual behaviour also contributes to overall environmental impact. People may recognize that certain activities are environmentally harmful but may not understand how much it damages the environment.

Food is mostly important because environmental sustainability and food security are closely connected. Food systems require land, water, energy, transportation, processing, and other resources.^[2] The sustainability of these systems is therefore relevant to SDG 2 - Zero Hunger, which includes promoting sustainable agriculture and sustainable food production systems. Food waste only increases environmental impact because the resources used to produce, process and transport discarded food are effectively wasted.

College students make decisions regarding food, transportation, electricity, and consumption, making them a suitable target group for an awareness-based solution. This project addresses this issue through a web-based carbon footprint calculator that allows students to estimate their emissions from different aspects of their lifestyle. Rather than providing only a single numerical value, the system presents a category-wise breakdown, identifies the major contributors, and provides practical recommendations for reducing environmental impact.

This solution therefore connects individual carbon-footprint awareness with sustainable food consumption, providing an accessible approach that supports the environmental objectives associated with SDG 2 while encouraging more responsible consumption.

PROBLEM STATEMENT

High carbon emissions are associated with different everyday activities, but people often lack a simple method for measuring the environmental impact of their own lifestyle.

College students may use private transportation regularly, consume electricity, purchase food, waste leftovers and dispose of materials without understanding the associated carbon emissions.

Food is more important because it connects the environmental problem with **SDG 2 - Zero Hunger**. Food production requires many resources, and food that is unnecessarily wasted represents the loss of those resources as well as the emissions associated with producing and transporting the food.

Therefore, the problem addressed by this project is:

College students lack a simple and interactive method to estimate the carbon emissions associated with their everyday activities, particularly food consumption and food waste, and to identify practical ways of reducing their environmental impact.

The solution is a food-focused personal carbon footprint calculator that estimates emissions from multiple lifestyle categories, highlights food-related impacts, and provides recommendations for more sustainable behaviour.

SUSTAINABILITY ANALYSIS

SDG 2 - Zero Hunger^[1]

The primary Sustainable Development Goal associated with this project is **SDG 2 - Zero Hunger**.

SDG 2 aims to end hunger, achieve food security and improved nutrition, and promote sustainable agriculture. Target 2.4 addresses sustainable food production systems and good agricultural practices.

The calculator supports this aspect of SDG 2 by increasing awareness of the environmental impact of food consumption and food waste.

The project does not directly solve hunger. Instead, it brings awareness to a supporting issue: the sustainability of food consumption. Reducing food waste and encouraging responsible consumption can reduce unnecessary environmental pressure on food systems.

A sustainable food system must continue to provide food without causing excessive environmental degradation. Climate change and environmental pressures can affect agricultural productivity and food security. Therefore, promoting sustainable consumption contributes indirectly to the broader objectives of SDG 2.

SDG 12 - Responsible Consumption and Production^[3]

SDG 12 - Responsible Consumption and Production is also relevant to the system.

The objective of SDG 12 is to encourage more sustainable patterns of consumption and production. Food waste is particularly important because reducing food waste at the consumer level is one of the targets associated with SDG 12.

The calculator encourages users to see their consumption patterns and understand the consequences of unnecessary consumption and food waste.

SDG 13 - Climate Action^[4]

The project is also related to **SDG 13 - Climate Action** because its primary environmental problem is high carbon emissions.

The calculator helps users understand their estimated contribution to greenhouse-gas emissions from everyday activities.

By providing users with calculated information and practical recommendations, the system encourages awareness of climate-related impacts.

Therefore, the project has connections with:

- **SDG 2 – Zero Hunger:** Sustainable food systems
- **SDG 12 – Responsible Consumption and Production:** Reducing unnecessary consumption and food waste
- **SDG 13 – Climate Action:** Reducing awareness and behavioural drivers of carbon emissions SDG 2 remains the primary focus of the project.

ISO 14001^[5]

ISO 14001 provides a framework for environmental management systems. Although the calculator is not itself an ISO 14001 environmental management system, its approach is related to the principle of monitoring and improving environmental performance.

The system follows:

Measure ➡ Identify ➡ Act ➡ Improve

The user's activities are measured, the major sources of emissions are then identified, recommendations are suggested, and users can make changes based on the recommendations.

Therefore, ISO 14001 is relevant as a supporting environmental-management concept rather than as a direct requirement.

THE ENGINEERING SOLUTION

Overview

The solution is a **Food-Focused Personal Carbon Footprint Calculator for College Students**.

The system will be developed as an interactive web application that allows users to enter information about their everyday activities and receive an estimated carbon footprint.

The calculator will cover five main areas:

1. **Food and food waste**
2. **Transportation**
3. **Electricity**
4. **Waste**
5. **General consumption**

Food will receive the greatest emphasis because it provides the project's connection to SDG 2.

The system will follow four stages:

Measure ➡ Identify ➡ Act ➡ Improve

Target Users

The primary target users are college students. Students have different lifestyles and may live at home, in hostels or as Paying Guests(PGs). They also use different forms of transportation and have different food and consumption patterns. The system will therefore use simple questions that the users can answer without requiring much technical environmental knowledge.

Food and Food-Waste Component

Food is the main focus of this calculator because it provides the strongest connection to SDG 2. Users can enter information such as their general food consumption, frequency of eating outside, food purchasing habits and the approximate amount of food they have wasted.

The calculator uses this information to estimate the emissions associated with food consumption and waste. The purpose is not to judge the user's diet, but to show how their food-related habits contribute to the overall carbon footprint and how they can improve their lifestyles for the future.

Other Components

The calculator also considers **transportation, electricity, general waste and other consumption**. These components will work using the same basic principle as the food component: the user provides information about their activities, and the system uses appropriate emission factors to estimate the respective CO₂e emissions.

For example, transportation can be calculated using the distance travelled and mode of transport, while electricity can be estimated from the amount of electricity consumed. Waste and general consumption can similarly be included using suitable activity data and emission factors.

The basic calculation used by this system is:

Carbon Emissions = Activity Data × Emission Factor

For example:

Transportation emissions = Distance travelled × Transportation emission factor

Electricity emissions = Electricity consumed × Electricity emission factor

Similar calculations can be applied to food and waste categories.

The results will be expressed in **kg CO₂e**, allowing different greenhouse-gas emissions to be represented using a common unit.

The emission factors used in the final website will be obtained from reliable published sources.

System Workflow

The system can be represented as:

User Input



Input Validation



Activity Data



Emission Factors



Emission Calculation



Category-wise Results



Total Carbon Footprint



Major Contributor Identified



Recommendations

This workflow provides the technical basis of the application

.

Results and Recommendations

The results page will display the user's estimated carbon footprint.

For example:

Estimated Monthly Carbon Footprint: XXX kg CO₂e

A category-wise breakdown will then be displayed.

Category	Contribution
Food	XX%
Transportation	XX%
Electricity	XX%
Waste	XX%
Others	XX%

Charts can be used to make the results easier to understand.

The system will also identify the largest contributor.

For example:

Food is your largest estimated source of emissions.

The system can then provide recommendations based on the user's results.

For food:

- Reduce avoidable food waste.
- Plan purchases according to actual requirements.
- Store food properly.
- Use leftovers where practical.
- Consider lower-impact food choices.

The same can be done for the other categories aswell. The goal is to provide **practical suggestions rather than forcing a particular lifestyle.**

FEASIBILITY ANALYSIS

Technical Feasibility

This system is technically feasible using standard web technologies.

The core application can be developed using:

- **HTML** – webpage structure
- **CSS** – layout and visual design
- **JavaScript** – user interaction and calculations

The calculation engine does not require any advanced technology. The system can operate using predefined emission factors and mathematical calculations taken from credible sources, making implementation manageable.

Economic Feasibility

This system has low development costs because it does not require specialized hardware. The application can be developed using commonly available software development tools and can be hosted using a low-cost or free hosting service for demonstration purposes.

Operational Feasibility

This system is designed for easy use. Users can provide information about familiar activities such as food consumption, travel, electricity and waste. The system then processes the information and displays the results visually. This simple interface makes it suitable for college students without requiring special environmental knowledge.

Environmental Feasibility

The software itself does not directly remove carbon emissions.

Its environmental benefit is indirect:

Awareness → Understanding → Behavioural Change → Reduced Waste/Consumption → Potentially Lower Emissions

The strongest expected impact is through increased awareness of food waste and consumption patterns.

EXPECTED ENVIRONMENTAL BENEFITS

This system is expected to provide the following benefits.

Increased Awareness

Students can see an estimate of the environmental impact associated with their own lifestyle.

Identification of Major Contributors

The category-wise breakdown allows users to identify which activities contribute most to their estimated footprint.

Reduced Food Waste

By specifically measuring food waste, the system encourages users to recognize and reduce avoidable food disposal.

Responsible Consumption

Users may become more conscious of purchasing, transportation, electricity and food-related decisions.

Resource Conservation

Reducing unnecessary consumption and food waste can reduce the indirect use of resources such as land, water and energy.

Support for Sustainable Food Systems

Encouraging responsible food consumption and reducing avoidable food waste supports the broader objective of developing more sustainable food systems associated with SDG 2.

JUSTIFICATION OF THE SOLUTION

This solution is justified because it addresses high carbon emissions at the level of everyday individual behaviour.

Many people understand that transportation, electricity consumption and food waste have environmental consequences, but they may not know which activity contributes the most.

This calculator solves the problem by converting everyday activity information into an understandable carbon-footprint estimations.

The **food-focused design** provides the main point of distinction from a generic carbon calculator. Food consumption and food waste are emphasized because they connect the environmental problem with **SDG 2 - Zero Hunger and sustainable food systems**.

This system does not claim that a carbon calculator can directly eliminate hunger. Instead, it supports the sustainability aspect of SDG 2 by encouraging more responsible use of food resources and reducing food waste.

This solution is also suitable for college students because it is:

- Simple to use
- Interactive
- Low-cost
- Web-based
- Easy to understand
- Technically feasible

The use of HTML, CSS and JavaScript also keeps implementation within the capabilities of a student development team. Most importantly, the system does not stop at displaying a carbon-footprint number. It explains the result and gives users practical actions they can take.

This solution therefore provides a balance between **technical simplicity, environmental relevance, SDG alignment and practical usability**.

CONCLUSION

High carbon emissions are associated with many everyday activities, such as transportation, electricity consumption, food consumption and waste generation. Individual awareness is therefore an important supporting component of broader sustainability efforts.

This report proposed a **Food-Focused Personal Carbon Footprint Calculator for College Students** as a solution to this problem of high carbon emissions. The system estimates emissions from food, food waste, transportation, electricity, waste and general consumption. Food is given particular emphasis because of its relationship with sustainable food systems and **SDG 2 - Zero Hunger**.

This system follows a **Measure-Analyze-Educate-Recommend** approach. Users enter information about their lifestyle, receive an estimated carbon footprint, identify the major causes of emissions, and receive practical recommendations for reducing their environmental impact. The solution is technically and economically feasible because it can be developed using standard web technologies without requiring complex technologies or specialized hardware.

Although the calculator cannot directly solve hunger or climate change, it can increase awareness of the environmental consequences of individual behaviour. By highlighting food consumption and food waste, it can encourage more

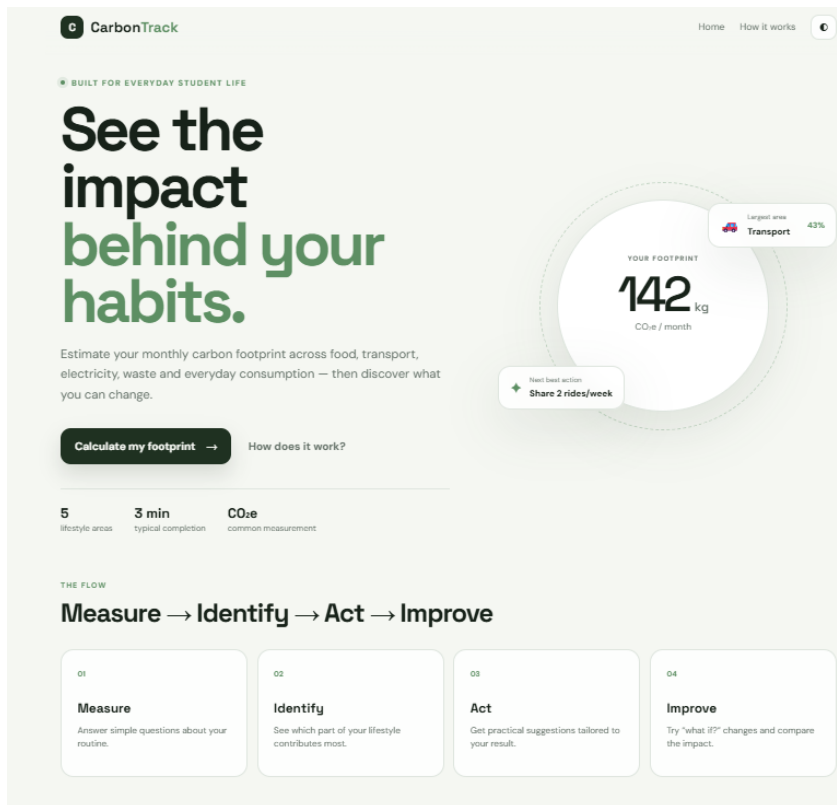
responsible use of food resources and support the goal of sustainable food systems. This calculator therefore provides a simple, interactive and feasible approach to promoting sustainable behaviour among college students while addressing the environmental problems of high carbon emissions.

REFERENCES

- [1] United Nations, “Goal 2: Zero Hunger,” *United Nations Sustainable Development Goals*. [Online]. Available: <https://www.un.org/sustainabledevelopment/hunger/>. [Accessed: Sep. 6, 2026].
- [2] Food and Agriculture Organization of the United Nations, “Greenhouse gas emissions from agrifood systems,” *FAO*. [Online]. Available: <https://www.fao.org/faostat/en/#data/GT>. [Accessed: Sep. 6, 2026].
- [3] United Nations, “Goal 12: Responsible Consumption and Production,” *United Nations Sustainable Development Goals*. [Online]. Available: <https://www.un.org/sustainabledevelopment/sustainable-consumption-production/>. [Accessed: Sep. 6, 2026].
- [4] United Nations, “Goal 13: Climate Action,” *United Nations Sustainable Development Goals*. [Online]. Available: <https://www.un.org/sustainabledevelopment/climate-change/>. [Accessed: Sep. 7, 2026].
- [5] International Organization for Standardization, “ISO 14001:2026 — Environmental management systems — Requirements with guidance for use,” *ISO*, 2026. [Online]. Available: <https://www.iso.org/standard/14001>. [Accessed: Sep. 7, 2026].

APPENDIX

Appendix A – Website Screenshots



CarbonTrack Home How it works

← Back Food & drink 1 / 5

Food & drink

Tell us about your usual food habits. Estimates are designed to be simple, not judgmental.

What best describes your usual diet?

☐ Mostly plant-based ☒ Mixed ☐ Mostly meat-based ☐ Mostly vegetarian

How often do you eat outside each week?

3x

How much food do you usually waste?

☐ Almost none ☒ A little ☐ Frequently

Food deliveries per week

2x

Continue →

[Home](#)
[How it works](#)

← Back

Transportation

2 / 5

CATEGORY 02

Transportation

Your daily commute and other travel can make a significant difference to your footprint.

How do you usually travel to college?

Motorcycle / scooter

Car

Bus

Metro / train

Auto-rickshaw

Walk / cycle

Approximate one-way distance

12 km

Days per week you make this commute

5 days

Do you regularly reduce the impact of these trips?

Not usually

Carpool / share rides

Switch some trips to public transport

Previous

Continue →

[Home](#)
[How it works](#)

← Back

Electricity

3 / 5

CATEGORY 03

Electricity

Use your approximate monthly electricity bill, or let us make a simple estimate.

How do you want to estimate electricity?

I know my monthly kWh

Estimate from my devices

Monthly electricity usage (kWh)

100

Approximate is fine. Check your electricity bill if available.

If estimating: how many regularly used devices/appliances?

4

Previous

Continue →

CarbonTrack

Home

How it works

← Back

Waste

4 / 5

CATEGORY 04

Waste

Think about how much waste you generate and what happens to recyclable material.

How would you describe your usual waste generation?

Low

Moderate

High

Do you separate recyclable material?

Usually

Sometimes

Not usually

Previous

Continue →

← Back

Consumption

5 / 5

CATEGORY 05

Everyday consumption

A simple snapshot of clothing, electronics and other purchases.

How often do you buy new clothes?

Rarely

Every few months

Frequently

How often do you buy electronics?

Rarely

Occasionally

Frequently

How often do you make online purchases?

Rarely

Sometimes

Often

Previous

Calculate footprint →

YOUR MONTHLY ESTIMATE

Here's where your footprint comes from.

Start again

ESTIMATED TOTAL

306
kg CO₂e
per month

This is an estimate based on the activity information you entered and the emission factors configured for the calculator.

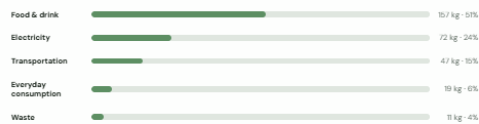
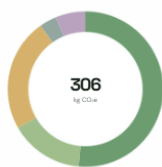
BIGGEST CONTRIBUTOR

**157 kg****Food & drink**

51% of your estimated monthly footprint comes from food & drink.

BREAKDOWN

What makes up your footprint?



YOUR NEXT MOVE

YOUR NEXT MOVE

Start by reducing avoidable food waste.

FOOD & DRINK

157 kg CO₂e/month

Food has the largest estimated share of your footprint. Focus first on waste and convenience habits rather than trying to completely change your diet overnight.

1 Plan purchases

Buy closer to what you realistically use.

2 Use leftovers

Keep edible food from becoming unnecessary waste.

3 Store food properly

Better storage can extend shelf life and prevent avoidable disposal.

01

Plan purchases

Buy closer to what you realistically use.

02

Use leftovers

Keep edible food from becoming unnecessary waste.

WHAT IF?

Experiment with your habits.

See an example of how a single change could affect your estimate.



Make one 15% cut

Reduce your biggest category by 15%.

24 kg less / month



Reduce waste

Cut your waste-related impact by 50%.

3 kg less / month



Combine two habits

Reduce your two largest categories by 10%.

23 kg less / month

Illustrative footprint after selected change

282 kg CO₂e

KEEP GOING

Your footprint is a starting point, not a verdict.

Small, repeatable changes are easier to maintain than trying to change everything at once.

Recalculate

Appendix B – HTML,JS and CSS code

index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <meta name="description" content="CarbonTrack - a student-friendly personal carbon footprint calculator." />

  <title>CarbonTrack — Personal Carbon Footprint</title>

  <link rel="preconnect" href="https://fonts.googleapis.com">

  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

  <link href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family=Space+Grotesk:wght@500;600;700&display=swap" rel="stylesheet">

  <link rel="stylesheet" href="css/style.css" />

</head>

<body>

  <div class="ambient ambient-one"></div>

  <div class="ambient ambient-two"></div>

  <header class="site-header">

    <a href="#home" class="brand" data-nav="home">

      <span class="brand-mark">C</span>

      <span>Carbon<span>Track</span></span></span>

    </a>

    <nav>

      <a href="#home" data-nav="home">Home</a>

      <a href="#how" data-nav="how">How it works</a>

      <button class="theme-btn" id="themeToggle" aria-label="Toggle theme">☐</button>

    </nav>
```

```
</header>
```

```
<main>
```

```
<section id="home" class="screen active">
```

```
<div class="hero">
```

```
<div class="hero-copy">
```

```
<div class="eyebrow"><span class="pulse"></span> Built for everyday student  
life</div>
```

```
<h1>See the impact<br><em>behind your habits.</em></h1>
```

```
<p class="hero-text">
```

Estimate your monthly carbon footprint across food, transport, electricity, waste and everyday consumption — then discover what you can change.

```
</p>
```

```
<div class="hero-actions">
```

```
<button class="primary-btn" id="startBtn">Calculate my footprint  
<span>→</span></button>
```

```
<button class="text-btn" id="howBtn">How does it work?</button>
```

```
</div>
```

```
<div class="micro-stats">
```

```
<div><strong>5</strong><span>lifestyle areas</span></div>
```

```
<div><strong>3 min</strong><span>typical completion</span></div>
```

```
<div><strong>CO2e</strong><span>common measurement</span></div>
```

```
</div>
```

```
</div>
```

```
<div class="hero-visual">
```

```
<div class="orbit-card orbit-main">
```

```
<div class="mini-label">YOUR FOOTPRINT</div>
```

```
<div class="orbit-number">142<span>kg</span></div>
```

```
<div class="orbit-sub">CO2e / month</div>
```

```
<div class="mini-bars">
```

```
<span style="width:72%"></span><span style="width:48%"></span><span  
style="width:38%"></span><span style="width:23%"></span>
```

```

    </div>

    </div>

    <div class="floating-card category-card"><span> 🚗 </span><div><small>Largest
area</small><strong>Transport</strong></div><b>43%</b></div>

    <div class="floating-card tip-card"><span class="tip-
icon"> ♦ </span><div><small>Next best action</small><strong>Share 2
rides/week</strong></div></div>

    </div>

</div>

<div id="how" class="how-strip">

    <div class="section-heading"><span>THE FLOW</span><h2>Measure → Identify →
Act → Improve</h2></div>

    <div class="flow-grid">

        <article><span>01</span><h3>Measure</h3><p>Answer simple questions about your
routine.</p></article>

        <article><span>02</span><h3>Identify</h3><p>See which part of your lifestyle
contributes most.</p></article>

        <article><span>03</span><h3>Act</h3><p>Get practical suggestions tailored to your
result.</p></article>

        <article><span>04</span><h3>Improve</h3><p>Try “what if?” changes and compare
the impact.</p></article>

    </div>

</div>

</section>

<section id="calculator" class="screen app-screen">

    <div class="calculator-shell">

        <div class="calc-top">

            <button class="back-btn" id="backBtn">← Back</button>

            <div class="progress-wrap">

                <div class="progress-meta"><span id="stepLabel">Food & drink</span><span
id="stepCount">1 / 5</span></div>

```

```
<div class="progress-track"><div class="progress-bar"
id="progressBar"></div></div>
```

```
</div>
```

```
</div>
```

```
<div class="question-card" id="questionCard">
```

```
<div class="question-icon" id="questionIcon">❓</div>
```

```
<div class="question-copy">
```

```
<div class="section-kicker" id="questionKicker">CATEGORY 01</div>
```

```
<h2 id="questionTitle">Food & drink</h2>
```

```
<p id="questionDesc">Tell us about your usual food habits. Estimates are designed to
be simple, not judgmental.</p>
```

```
</div>
```

```
<div class="question-body" id="questionBody"></div>
```

```
</div>
```

```
<div class="calc-actions">
```

```
<button class="secondary-btn" id="prevStepBtn">Previous</button>
```

```
<button class="primary-btn" id="nextStepBtn">Continue <span>→</span></button>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<section id="results" class="screen">
```

```
<div class="results-shell">
```

```
<div class="results-header">
```

```
<div>
```

```
<div class="eyebrow"><span class="pulse"></span> Your monthly estimate</div>
```

```
<h2>Here's where your<br><em>footprint comes from.</em></h2>
```

```
</div>
```

```
<button class="secondary-btn" id="restartTop">Start again</button>
```

</div>

<div class="result-hero">

<div class="score-card">

<div class="score-label">ESTIMATED TOTAL</div>

<div class="score-value">0<small>kg
CO₂e</small></div>

<div class="score-period">per month</div>

<div class="score-note" id="scoreNote">Based on the lifestyle information you
entered.</div>

</div>

<div class="biggest-card">

<div class="mini-label">BIGGEST CONTRIBUTOR</div>

<div class="biggest-icon" id="biggestIcon"> 🚗 </div>

<h3 id="biggestName">Transportation</h3>

<div class="biggest-number">0 kg</div>

<p id="biggestText">This category is your largest estimated source of emissions.</p>

</div>

</div>

<div class="breakdown-section">

<div class="section-heading">BREAKDOWN<h2>What makes up
your footprint?</h2></div>

<div class="breakdown-grid">

<div class="chart-card"><canvas id="donutChart" width="320"
height="320"></canvas><div class="chart-center"><strong
id="chartCenter">0kg CO₂e</div></div>

<div class="bars-card" id="breakdownBars"></div>

</div>

</div>

<div class="recommendation-section">

```

    <div class="section-heading"><span>YOUR NEXT MOVE</span><h2
id="recTitle">Start with your biggest contributor.</h2></div>

    <div class="rec-layout">

        <div class="rec-main" id="recMain"></div>

        <div class="rec-side">

            <div class="tiny-card"><span>01</span><strong id="recSmall1">Make one
change</strong><p id="recSmallText1">Pick an action that feels realistic this
week.</p></div>

            <div class="tiny-card">

                <span>02</span>

                <strong id="recSmall2">Second recommendation</strong>

                <p id="recSmallText2">Your second recommended action will appear here.</p>
</div>

            </div>

        </div>

    </div>

    <div class="whatif-section">

        <div class="section-heading"><span>WHAT IF?</span><h2>Experiment with your
habits.</h2><p>See an example of how a single change could affect your
estimate.</p></div>

        <div class="whatif-grid" id="whatIfGrid"></div>

        <div class="whatif-total"><span>Illustrative footprint after selected
change</span><strong><span id="whatIfTotal">0</span> kg CO2e</strong></div>

        </div>

    <div class="final-cta">

        <div><span class="eyebrow">KEEP GOING</span><h2>Your footprint is a starting
point, not a verdict.</h2><p>Small, repeatable changes are easier to maintain than trying to
change everything at once.</p></div>

        <button class="primary-btn" id="restartBottom">Recalculate
<span>↺</span></button>

    </div>

```

```
<footer class="site-footer">
  <span>CarbonTrack • Student carbon footprint calculator</span>
  <span>Estimates are educational and depend on the emission factors used.</span>
</footer>
</div>
</section>
</main>

<script src="js/emissions.js"></script>
<script src="js/app.js"></script>
</body>
</html>
```

app.js

```
const state = {
  step: 0,
  answers: {
    foodType: "mixed",
    foodWaste: "wasteMedium",
    foodOut: 3,
    delivery: 2,
    transportMode: "bike",
    transportDistance: 12,
    transportDays: 5,
    transportExtra: "none",
    electricityMode: "direct",
    electricityKwh: 100,
    devices: 4,
    wasteLevel: "medium",
    recycle: "sometimes",
```

```

    clothing: "Medium",
    electronics: "Medium",
    online: "Medium"
  },
  result: null,
  whatIfReduction: 0
};

const steps = [
  {
    key: "food",
    label: "Food & drink",
    icon: "🍷",
    title: "Food & drink",
    desc: "Tell us about your usual food habits. Estimates are designed to be simple, not judgmental.",
    render: renderFood
  },
  {
    key: "transport",
    label: "Transportation",
    icon: "🚗",
    title: "Transportation",
    desc: "Your daily commute and other travel can make a significant difference to your footprint.",
    render: renderTransport
  },
  {
    key: "electricity",
    label: "Electricity",
    icon: "⚡",

```

```

    title: "Electricity",
    desc: "Use your approximate monthly electricity bill, or let us make a simple estimate.",
    render: renderElectricity
  },
  {
    key: "waste",
    label: "Waste",
    icon: "♻️",
    title: "Waste",
    desc: "Think about how much waste you generate and what happens to recyclable material.",
    render: renderWaste
  },
  {
    key: "consumption",
    label: "Consumption",
    icon: "🛒",
    title: "Everyday consumption",
    desc: "A simple snapshot of clothing, electronics and other purchases.",
    render: renderConsumption
  }
];

```

```
const $ = (id) => document.getElementById(id);
```

```

function showScreen(id){
  document.querySelectorAll(".screen").forEach(s => s.classList.remove("active"));
  $(id).classList.add("active");
  window.scrollTo({top:0, behavior:"smooth"});
}

```

```

function startCalculator(){
  state.step = 0;
  renderStep();
  showScreen("calculator");
}

function renderStep(){
  const step = steps[state.step];
  $("stepLabel").textContent = step.label;
  $("stepCount").textContent = `${state.step + 1} / ${steps.length}`;
  $("progressBar").style.width = `${((state.step + 1) / steps.length) * 100}%`;
  $("questionIcon").textContent = step.icon;
  $("questionKicker").textContent = `CATEGORY 0${state.step + 1}`;
  $("questionTitle").textContent = step.title;
  $("questionDesc").textContent = step.desc;
  $("questionBody").innerHTML = step.render();
  $("prevStepBtn").style.visibility = state.step === 0 ? "hidden" : "visible";
  $("nextStepBtn").innerHTML = state.step === steps.length - 1 ? "Calculate footprint"
  <span>→</span> : "Continue <span>→</span>";
  wireInputs();
}

function renderFood(){
  return `
    ${choiceGroup("What best describes your usual diet?", "foodType", [
      ["plant", "Mostly plant-based"], ["mixed", "Mixed"], ["meat", "Mostly meat-
based"], ["plant", "Mostly vegetarian"]
    ])}
    <div class="form-group">
      <label class="form-label">How often do you eat outside each week?</label>

```

```

    <div class="range-wrap"><input data-range="foodOut" type="range" min="0" max="14"
step="1" value="{state.answers.foodOut}"><span class="range-value"
id="foodOutValue">{state.answers.foodOut} ×</span></div>

```

```

</div>

```

```

    {choiceGroup("How much food do you usually waste?", "foodWaste", [
        ["wasteLow", "Almost none"], ["wasteMedium", "A little"], ["wasteHigh", "Frequently"]
    ])}

```

```

<div class="form-group">

```

```

    <label class="form-label">Food deliveries per week</label>

```

```

    <div class="range-wrap"><input data-range="delivery" type="range" min="0" max="14"
step="1" value="{state.answers.delivery}"><span class="range-value"
id="deliveryValue">{state.answers.delivery} ×</span></div>

```

```

</div>`;

```

```

}

```

```

function renderTransport() {

```

```

    return `

```

```

        {choiceGroup("How do you usually travel to college?", "transportMode", [
            ["bike", "Motorcycle / scooter"], ["car", "Car"], ["bus", "Bus"], ["metro", "Metro /
train"], ["auto", "Auto-rickshaw"], ["walk", "Walk / cycle"]

```

```

        ])}

```

```

    <div class="form-group">

```

```

        <label class="form-label">Approximate one-way distance</label>

```

```

        <div class="range-wrap"><input data-range="transportDistance" type="range" min="0"
max="60" step="1" value="{state.answers.transportDistance}"><span class="range-value"
id="transportDistanceValue">{state.answers.transportDistance} km</span></div>

```

```

    </div>

```

```

    <div class="form-group">

```

```

        <label class="form-label">Days per week you make this commute</label>

```

```

        <div class="range-wrap"><input data-range="transportDays" type="range" min="0"
max="7" step="1" value="{state.answers.transportDays}"><span class="range-value"
id="transportDaysValue">{state.answers.transportDays} days</span></div>

```

```

    </div>

```

```

    {choiceGroup("Do you regularly reduce the impact of these trips?", "transportExtra", [

```

```
["none","Not usually"],["carpool","Carpool / share rides"],["public","Switch some trips to public transport"]
```

```
  ]} `;
```

```
}
```

```
function renderElectricity(){
```

```
  return `
```

```
    ${choiceGroup("How do you want to estimate electricity?", "electricityMode", [
```

```
      ["direct","I know my monthly kWh"],["estimate","Estimate from my devices"]
```

```
    ])}  
    <div class="form-group">
```

```
      <label class="form-label">Monthly electricity usage (kWh)</label>
```

```
      <input id="electricityKwhInput" type="number" min="0" max="2000" step="1" value="${state.answers.electricityKwh}" ${state.answers.electricityMode==="estimate" ? "disabled" : ""}>
```

```
      <small style="color:var(--muted);display:block;margin-top:7px">Approximate is fine. Check your electricity bill if available.</small>
```

```
    </div>
```

```
    <div class="form-group">
```

```
      <label class="form-label">If estimating: how many regularly used devices/appliances?</label>
```

```
      <div class="range-wrap"><input data-range="devices" type="range" min="1" max="12" step="1" value="${state.answers.devices}"><span class="range-value" id="devicesValue">${state.answers.devices}</span></div>
```

```
    </div> `;
```

```
}
```

```
function renderWaste(){
```

```
  return `
```

```
    ${choiceGroup("How would you describe your usual waste generation?", "wasteLevel", [
```

```
      ["low","Low"],["medium","Moderate"],["high","High"]
```

```
    ])}  
    ${choiceGroup("Do you separate recyclable material?", "recycle", [
```

```

    ["yes", "Usually"], ["sometimes", "Sometimes"], ["no", "Not usually"]
  ]} `;
}

```

```

function renderConsumption(){
  return `
    ${choiceGroup("How often do you buy new clothes?", "clothing", [
      ["Low", "Rarely"], ["Medium", "Every few months"], ["High", "Frequently"]
    ])}
    ${choiceGroup("How often do you buy electronics?", "electronics", [
      ["Low", "Rarely"], ["Medium", "Occasionally"], ["High", "Frequently"]
    ])}
    ${choiceGroup("How often do you make online purchases?", "online", [
      ["Low", "Rarely"], ["Medium", "Sometimes"], ["High", "Often"]
    ])} `;
}

```

```

function choiceGroup(label, name, options){
  return `<div class="form-group"><label class="form-label">${label}</label><div
class="option-grid">${options.map(([value, text], index) => {
    const id = `${name}-${index}`;
    return `<div class="option"><input id="${id}" type="radio" name="${name}"
value="${value}" ${String(state.answers[name]) === String(value) ? "checked":""}><label
for="${id}">${text}</label></div>`;
  }).join("")}</div></div>`;
}

```

```

function wireInputs(){
  document.querySelectorAll("#questionBody .option").forEach(option => {
    option.addEventListener("click", e => {
      if(e.target.tagName.toLowerCase() !== "input"){

```

```

const input = option.querySelector('input[type="radio"]');
if(input){
  input.checked = true;
  input.dispatchEvent(new Event("change", {bubbles:true}));
}
}
});
});
document.querySelectorAll("#questionBody input[type=radio]").forEach(input=>{
  input.addEventListener("change", e => {
    state.answers[e.target.name] = e.target.value;
    if(e.target.name === "electricityMode"){
      $("questionBody").innerHTML = steps[state.step].render();
      wireInputs();
    }
  });
});
document.querySelectorAll("[data-range]").forEach(input=>{
  const key = input.dataset.range;
  const valueId = `${key}Value`;
  input.addEventListener("input", e=>{
    const val = Number(e.target.value);
    state.answers[key]=val;
    if($(valueId)){
      const suffix = key==="transportDistance" ? " km" : key==="transportDays" ? " days" :
"×";
      $(valueId).textContent = val + suffix;
    }
  });
});
const electricityInput = $("electricityKwhInput");

```

```

    if(electricityInput){
      electricityInput.addEventListener("input", e => state.answers.electricityKwh =
        Number(e.target.value));
    }
  }
}

```

```

function calculateFootprint(a){
  const f = EMISSION_FACTORS;

  // Food: existing UI uses broad monthly proxy values.
  const foodBase = f.food[a.foodType] || f.food.mixed;
  const foodWaste = f.food[a.foodWaste] || 0;
  const foodOutside = (Number(a.foodOut) || 0) * 4.33 * f.food.eatingOut;
  const delivery = (Number(a.delivery) || 0) * 4.33 * f.food.delivery;
  const food = foodBase + foodWaste + foodOutside + delivery;

  // Transport: distance is one-way, so multiply by 2 for a return trip.
  // 4.33 converts weekly commuting into an approximate monthly value.
  const distance = Number(a.transportDistance) || 0;
  const days = Number(a.transportDays) || 0;
  const monthlyKm = distance * 2 * days * 4.33;
  let transport = monthlyKm * (f.transport[a.transportMode] || 0);

  if(a.transportExtra === "carpool"){
    transport *= 0.75;
  } else if(a.transportExtra === "public"){
    transport *= 0.85;
  }

  // Electricity: direct monthly kWh or a simple device estimate.

```

```

let kwh = Number(a.electricityKwh) || 0;
if(a.electricityMode === "estimate"){
  kwh = (Number(a.devices) || 0) * 8 * 30 / 1000;
}
const electricity = kwh * f.electricity;

// Waste and consumption remain transparent screening proxies.
let waste = f.waste[a.wasteLevel] || f.waste.medium;
if(a.recycle === "yes") waste *= f.waste.recycleBonus;
else if(a.recycle === "sometimes") waste *= 0.91;

const clothing = f.consumption["clothing" + a.clothing] || 0;
const electronics = f.consumption["electronics" + a.electronics] || 0;
const online = f.consumption["online" + a.online] || 0;
const consumption = clothing + electronics + online;

const categories = {food, transport, electricity, waste, consumption};
const total = Object.values(categories).reduce((sum, value) => sum + value, 0);
return {categories, total};
}

function nextStep(){
  if(state.step < steps.length - 1){
    state.step++;
    renderStep();
  } else {
    state.result = calculateFootprint(state.answers);
    renderResults();
    showScreen("results");
  }
}

```

```
}
```

```
function prevStep(){  
  if(state.step>0){state.step--;renderStep();}  
}
```

```
const CATEGORY_META = {  
  food: {name:"Food & drink", icon:"🍷"},  
  transport: {name:"Transportation", icon:"🚗"},  
  electricity: {name:"Electricity", icon:"⚡"},  
  waste: {name:"Waste", icon:"♻️"},  
  consumption: {name:"Everyday consumption", icon:"🛒"},  
};
```

```
function categoryPercentage(value,total){return total ? (value/total*100) : 0}
```

```
function renderResults(){  
  const {categories,total} = state.result;  
  $("totalFootprint").textContent = Math.round(total);  
  $("chartCenter").textContent = Math.round(total);  
  
  const ranked = Object.entries(categories).sort((a,b)=>b[1]-a[1]);  
  const [topKey, topAmount] = ranked[0];  
  $("biggestIcon").textContent = CATEGORY_META[topKey].icon;  
  $("biggestName").textContent = CATEGORY_META[topKey].name;  
  $("biggestAmount").textContent = Math.round(topAmount);  
  $("biggestText").textContent = `${Math.round(categoryPercentage(topAmount,total))}% of  
your estimated monthly footprint comes from  
${CATEGORY_META[topKey].name.toLowerCase()}.`;
```

```
$("#scoreNote").textContent = "This is an estimate based on the activity information you entered and the emission factors configured for the calculator.";
```

```
renderBars(categories,total);  
drawDonut(categories,total);  
renderRecommendations(topKey, topAmount, total);  
renderWhatIf(topKey, total);  
}
```

```
function renderBars(categories,total){  
  const sorted = Object.entries(categories).sort((a,b)=>b[1]-a[1]);  
  $("#breakdownBars").innerHTML = sorted.map(([key,value])=>  
    <div class="bar-row">  
      <div class="bar-label">${CATEGORY_META[key].name}</div>  
      <div class="bar-track"><div class="bar-fill"  
style="width:${categoryPercentage(value,total)}%"></div></div>  
      <div class="bar-number">${Math.round(value)} kg ·  
      ${Math.round(categoryPercentage(value,total))}%</div>  
    </div>`),join("");  
}
```

```
function drawDonut(categories,total){  
  const canvas = $("#donutChart"), ctx = canvas.getContext("2d");  
  const ratio = window.devicePixelRatio || 1;  
  const size = 320;  
  canvas.width = size*ratio; canvas.height = size*ratio; ctx.scale(ratio,ratio);  
  ctx.clearRect(0,0,size,size);  
  const entries = Object.entries(categories);  
  const colors = ["#6c9c70","#9fbe8c","#d5b16c","#93a6a0","#b8a2be"];  
  let start = -Math.PI/2;  
  entries.forEach(([key,value],i)=>{
```

```

const angle = total ? (value/total)*Math.PI*2 : 0;
ctx.beginPath();
ctx.arc(size/2,size/2,115,start,start+angle);
ctx.lineWidth=35;
ctx.strokeStyle=colors[i];
ctx.lineCap="butt";
ctx.stroke();
start += angle;
});
// center mask
ctx.beginPath(); ctx.arc(size/2,size/2,92,0,Math.PI*2);
ctx.fillStyle=getComputedStyle(document.body).getPropertyValue("--surface-solid");
ctx.fill();
}

const recommendations = {
  food: {
    title:"Start by reducing avoidable food waste.",
    tag:"FOOD & DRINK",
    main:"Food has the largest estimated share of your footprint. Focus first on waste and
convenience habits rather than trying to completely change your diet overnight.",
    items:[
      ["Plan purchases","Buy closer to what you realistically use."],
      ["Use leftovers","Keep edible food from becoming unnecessary waste."],
      ["Store food properly","Better storage can extend shelf life and prevent avoidable
disposal."]
    ]
  },
  transport: {
    title:"Make your commute your first lever.",
    tag:"TRANSPORTATION",

```

main:"Transportation is your largest estimated category. A few repeated changes to your normal commute can add up more than a one-off “perfect” choice.",

items:[

["Share two rides","Carpool or share a journey where practical."],

["Combine trips","Avoid separate journeys when one trip can handle several tasks."],

["Try public transport","Replace some individual journeys with bus or rail."]

]

},

electricity:{

title:"Trim the electricity you use most often.",

tag:"ELECTRICITY",

main:"Electricity is your largest estimated category. Target the appliances and habits you use for the longest periods rather than worrying about tiny one-off loads.",

items:[

["Reduce idle use","Switch off devices and appliances that are not needed."],

["Target long-running loads","Fans, AC and other long-use appliances deserve attention first."],

["Track your bill","Comparing monthly kWh makes improvement measurable."]

]

},

waste:{

title:"Reduce the amount that becomes waste.",

tag:"WASTE",

main:"Your waste category stands out in the estimate. Prevention is usually more useful than trying to manage waste after it has already been created.",

items:[

["Separate recyclables","Keep recoverable material out of general waste."],

["Avoid disposable items","Prefer reusables where practical."],

["Watch food waste","Unnecessary food disposal carries impacts from the resources used to produce it."]

]

},

```

consumption:{
  title:"Buy less, but buy more intentionally.",
  tag:"EVERYDAY CONSUMPTION",
  main:"Your everyday purchases add up. Focus on extending the life of what you already own and slowing down non-essential buying.",
  items:[
    ["Delay non-essential purchases","A short pause can reduce impulse buying."],
    ["Use what you own","Repair, reuse and keep items longer when practical."],
    ["Think about need","Choose purchases for usefulness, not just convenience."]
  ]
}
};

```

```

function renderRecommendations(key,amount,total){
  const r = recommendations[key];
  $("recTitle").textContent = r.title;
  $("recMain").innerHTML = `<div class="rec-
tag">${r.tag}</div><h3>${Math.round(amount)} kg
CO2e/month</h3><p>${r.main}</p><div class="rec-list">${r.items.map((item,i)=>`<div
class="rec-
item"><span>${i+1}</span><div><strong>${item[0]}</strong><p>${item[1]}</p></div></
div>`).join("")}</div>`;
  $("recSmall1").textContent = r.items[0][0];
  $("recSmallText1").textContent = r.items[0][1];
  $("recSmall2").textContent = r.items[1][0];
  $("recSmallText2").textContent = r.items[1][1];
}

```

```

function renderWhatIf(topKey,total){
  const options = [
    {icon:"↘",title:"Make one 15% cut",text:"Reduce your biggest category by
15%.",factor:.85},

```

```

    {icon:"♯",title:"Reduce waste",text:"Cut your waste-related impact by
30%.",special:"waste"},

    {icon:"♦",title:"Combine two habits",text:"Reduce your two largest categories by
10%.",factor2:.9}

];

$("whatIfGrid").innerHTML = options.map((o,i)=>`<div class="whatif-card
${i===0?"selected":""}" data-index="${i}"><button><div class="whatif-
icon">${o.icon}</div><h3>${o.title}</h3><p>${o.text}</p><div class="whatif-saving"
id="saving${i}"></div></button></div>`).join("");

const calc = () => {

  const cards = document.querySelectorAll(".whatif-card");

  cards.forEach(c=>c.classList.remove("selected"));

  const selected = [...cards].find(c=>Number(c.dataset.index)===state.whatIfReduction);

  if(selected) selected.classList.add("selected");

  let value = total;

  if(state.whatIfReduction===0){

    value = total - state.result.categories[topKey]*.15;

  } else if(state.whatIfReduction===1){

    value = total - state.result.categories.waste*.30;

  } else {

    const sorted = Object.entries(state.result.categories).sort((a,b)=>b[1]-a[1]);

    value = total - (sorted[0][1]+sorted[1][1])*10;

  }

  $("whatIfTotal").textContent = Math.max(0,Math.round(value));

  [...cards].forEach((c,i)=>{

    const saving = i===0 ? state.result.categories[topKey]*.15 : i===1 ?
state.result.categories.waste*.30 : Object.entries(state.result.categories).sort((a,b)=>b[1]-
a[1]).slice(0,2).reduce((a,[v])=>a+v,0)*.10;

    $('`saving${i}`').textContent = `≈ ${Math.round(saving)} kg less / month`;

  });

};

```

```

    document.querySelectorAll(".whatif-
card").forEach(c=>c.addEventListener("click",()=>{state.whatIfReduction=Number(c.dataset
.index);calc()});
    calc();
}

```

```

function restart(){showScreen("home");}

```

```

$("startBtn").addEventListener("click", startCalculator);
$("howBtn").addEventListener("click",
()=>document.querySelector("#how").scrollIntoView({behavior:"smooth"}));
$("nextStepBtn").addEventListener("click", nextStep);
$("prevStepBtn").addEventListener("click", prevStep);
$("backBtn").addEventListener("click", restart);
$("restartTop").addEventListener("click", restart);
$("restartBottom").addEventListener("click", restart);
$("themeToggle").addEventListener("click",()=>document.body.classList.toggle("dark"));

```

```

window.addEventListener("resize",()=>{if(state.result)
drawDonut(state.result.categories,state.result.total)});

```

```

showScreen("home");

```

emissions.js

```

/*

```

CarbonTrack emission-factor notes

The calculator uses transparent screening factors. They are not a certified inventory and should be reviewed with your supervisor before formal publication.

Electricity:

0.72 kg CO₂e/kWh is retained as a provisional grid-screening value.

Replace it with the India-specific grid factor selected from an official source for your reporting year.

Transport:

car 0.18 kg CO₂e/km
bike 0.09 kg CO₂e/km
bus 0.05 kg CO₂e/km
metro 0.035 kg CO₂e/km
auto 0.11 kg CO₂e/km
walk 0

These are simplified distance-based screening values. A formal calculator should instead use vehicle/fuel type, fuel economy, occupancy and a cited conversion-factor table.

Food:

The current interface asks for a diet pattern rather than food mass, so these are monthly screening proxies, not kg-per-kg food factors:
plant = 35, mixed = 55, meat = 85

Waste and consumption:

These are simplified screening scores because the interface does not collect measured kilograms of waste or purchase spending. They should not be described as universal physical emission factors.

Sources consulted:

- UK DESNZ, Greenhouse gas reporting: conversion factors 2026
- Our World in Data, Environmental Impacts of Food
- Poore and Nemecek (2018), as processed by Our World in Data

*/

```
const EMISSION_FACTORS = {  
  transport: {  
    car: 0.18,  
    bike: 0.09,  
    bus: 0.05,  
    metro: 0.035,  
    auto: 0.11,  
    walk: 0  
  },  
  electricity: 0.72,  
  food: {  
    plant: 35,  
    mixed: 55,  
    meat: 85,  
    wasteLow: 3,  
    wasteMedium: 7,  
    wasteHigh: 13,  
    eatingOut: 4,  
    delivery: 5  
  },  
  waste: {  
    low: 5,  
    medium: 12,  
    high: 21,  
    recycleBonus: 0.82  
  },  
  consumption: {  
    clothingLow: 3,  
    clothingMedium: 8,  
    clothingHigh: 16,  

```

```
electronicsLow: 2,  
electronicsMedium: 6,  
electronicsHigh: 12,  
onlineLow: 2,  
onlineMedium: 5,  
onlineHigh: 10  
}  
};
```

style.css

```
:root{  
  --bg:#f5f7f2;  
  --surface:rgba(255,255,255,.82);  
  --surface-solid:#ffffff;  
  --text:#172219;  
  --muted:#66736a;  
  --line:#dfe6df;  
  --accent:#5d8f63;  
  --accent-2:#c8dfc2;  
  --dark:#1f3121;  
  --shadow:0 25px 80px rgba(48,71,51,.12);  
  --radius:24px;  
}  
*{box-sizing:border-box}  
html{scroll-behavior:smooth}  
body{margin:0;font-family:"DM Sans",sans-serif;background:var(--bg);color:var(--text);min-height:100vh}  
body.dark{--bg:#101710;--surface:rgba(25,36,26,.9);--surface-solid:#1a261b;--text:#edf6ee;--muted:#a8b6aa;--line:#2b3a2c;--accent:#86b98b;--accent-2:#345136;--dark:#eaf4eb;--shadow:0 25px 80px rgba(0,0,0,.25)}  
button,input,select{font:inherit}
```

```

button {cursor:pointer}

.ambient {position:fixed;width:420px;height:420px;border-
radius:50%;filter:blur(90px);opacity:.25;pointer-events:none;z-index:-1}

.ambient-one {background:#bdddff;top:-140px;left:-110px}

.ambient-two {background:#dbe6b9;right:-150px;bottom:-100px}

.site-header {height:78px;display:flex;align-items:center;justify-content:space-between;max-
width:1180px;margin:auto;padding:0 22px;position:sticky;top:0;z-
index:30;background:linear-gradient(to bottom,var(--bg) 55%,transparent)}

.brand {display:flex;align-items:center;gap:10px;text-decoration:none;color:var(--text);font-
family:"Space Grotesk",sans-serif;font-size:21px;font-weight:700}

.brand>span:last-child span {color:var(--accent)}

.brand-mark {display:grid;place-items:center;width:34px;height:34px;border-
radius:11px;background:var(--dark);color:var(--bg);font-size:16px}

nav {display:flex;align-items:center;gap:22px}

nav a {color:var(--muted);text-decoration:none;font-size:14px}

.theme-btn {border:1px solid var(--line);background:var(--surface);color:var(--
text);width:38px;height:38px;border-radius:12px}

.screen {display:none;max-width:1180px;margin:auto;padding:35px 22px
80px;animation:fade .5s ease}

.screen.active {display:block}

@keyframes fade {from {opacity:0;transform:translateY(7px)} to {opacity:1;transform:none}}

.hero {min-height:590px;display:grid;grid-template-columns:1.06fr .94fr;align-
items:center;gap:55px}

.eyebrow {font-size:12px;letter-spacing:.14em;font-weight:700;color:var(--
accent);display:flex;align-items:center;gap:9px;text-transform:uppercase}

.pulse {width:7px;height:7px;background:var(--accent);border-radius:50%;box-shadow:0 0 0
5px rgba(93,143,99,.12)}

h1,h2,h3 {font-family:"Space Grotesk",sans-serif;margin:0}

h1 {font-size:clamp(58px,7vw,90px);line-height:.93;letter-spacing:-.055em;margin:20px 0}

h1 em,h2 em {font-style:normal;color:var(--accent)}

.hero-text {max-width:610px;color:var(--muted);font-size:18px;line-height:1.65}

.hero-actions {display:flex;align-items:center;gap:19px;margin:35px 0}

.primary-btn,.secondary-btn {border-radius:15px;padding:14px 19px;font-
weight:700;border:1px solid transparent;transition:.2s ease}

```

```

.primary-btn{background:var(--dark);color:var(--bg);box-shadow:0 14px 30px
rgba(31,49,33,.15)}

.primary-btn:hover{transform:translateY(-2px);box-shadow:0 17px 35px rgba(31,49,33,.2)}

.primary-btn span{margin-left:12px}

.secondary-btn{background:var(--surface);border-color:var(--line);color:var(--text)}

.secondary-btn:hover{transform:translateY(-1px);border-color:var(--accent)}

.text-btn,.back-btn{background:none;border:none;color:var(--muted);font-weight:700}

.micro-stats{display:flex;gap:32px;padding-top:23px;border-top:1px solid var(--line);max-
width:570px}

.micro-stats div{display:flex;flex-direction:column;gap:2px}

.micro-stats strong{font-family:"Space Grotesk";font-size:20px}

.micro-stats span{font-size:12px;color:var(--muted)}

.hero-visual{height:470px;position:relative;display:grid;place-items:center}

.hero-visual:before{content:"";position:absolute;width:360px;height:360px;border-
radius:50%;border:1px dashed #c6d8c7;transform:rotate(-18deg)}

.orbit-card{width:310px;height:310px;border-radius:50%;background:var(--surface-
solid);box-shadow:var(--shadow);display:flex;flex-direction:column;align-
items:center;justify-content:center;position:relative;z-index:2;border:1px solid var(--line)}

.mini-label,.score-label{font-size:10px;letter-spacing:.14em;font-weight:800;color:var(--
muted)}

.orbit-number{font-family:"Space Grotesk";font-size:72px;line-height:1;margin:10px 0
4px;letter-spacing:-.07em}

.orbit-number span{font-size:18px;letter-spacing:0;margin-left:8px;color:var(--muted)}

.orbit-sub{font-size:13px;color:var(--muted)}

.mini-bars{display:flex;gap:5px;margin-top:30px}

.mini-bars span{width:62px;height:5px;border-radius:99px;background:#b7d2ba}

.floating-card{position:absolute;background:var(--surface);backdrop-
filter:blur(14px);border:1px solid var(--line);box-shadow:var(--shadow);border-
radius:18px;padding:13px 16px;display:flex;align-items:center;gap:11px;z-index:4}

.floating-card small{display:block;color:var(--muted);font-size:10px;margin-bottom:2px}

.floating-card strong{font-size:13px}

.floating-card>span{font-size:22px}

.floating-card b{font-size:12px;color:var(--accent);margin-left:14px}

.category-card{right:0;top:85px}

```

```

.tip-card {left: 12px; bottom: 83px}

.tip-icon {color: var(--accent)}

.how-strip {padding: 68px 0 15px}

.section-heading > span {display: block; color: var(--accent); font-size: 11px; letter-spacing: .15em; font-weight: 800; margin-bottom: 10px}

.section-heading h2 {font-size: 38px; letter-spacing: -.04em}

.flow-grid {display: grid; grid-template-columns: repeat(4, 1fr); gap: 14px; margin-top: 28px}

.flow-grid article {padding: 23px; border: 1px solid var(--line); border-radius: 19px; background: var(--surface)}

.flow-grid article span {color: var(--accent); font-size: 11px; font-weight: 800}

.flow-grid h3 {font-size: 19px; margin-top: 28px}

.flow-grid p {color: var(--muted); font-size: 13px; line-height: 1.55}

.app-screen {max-width: none; padding-top: 15px}

.calculator-shell {max-width: 930px; margin: auto}

.calc-top {display: flex; align-items: center; gap: 25px; margin: 25px 0}

.progress-wrap {flex: 1; max-width: 650px; margin-left: auto}

.progress-meta {display: flex; justify-content: space-between; color: var(--muted); font-size: 12px; margin-bottom: 8px}

.progress-track {height: 7px; background: var(--line); border-radius: 99px; overflow: hidden}

.progress-bar {height: 100%; width: 20%; background: var(--accent); border-radius: 99px; transition: .35s ease}

.question-card {background: var(--surface); border: 1px solid var(--line); border-radius: 28px; padding: 42px; box-shadow: var(--shadow)}

.question-icon {width: 58px; height: 58px; border-radius: 17px; background: #e3eee2; display: grid; place-items: center; font-size: 27px; margin-bottom: 20px}

body.dark .question-icon {background: #2a412c}

.question-copy h2 {font-size: 42px; letter-spacing: -.045em}

.question-copy p {color: var(--muted); max-width: 660px; line-height: 1.6; margin: 12px 0 0}

.section-kicker {font-size: 11px; color: var(--accent); font-weight: 800; letter-spacing: .14em; margin-bottom: 8px}

.question-body {margin-top: 35px}

.form-group {margin-bottom: 27px}

```

```

.form-label{display:block;font-weight:700;font-size:14px;margin-bottom:10px}
.option-grid{display:grid;grid-template-columns:repeat(2,1fr);gap:10px}
.option{position:relative}
.option input{position:absolute;opacity:0;pointer-events:none}
.option label{display:block;border:1px solid var(--line);background:var(--surface-solid);padding:15px 16px;border-radius:15px;color:var(--text);transition:.2s}
.option label:hover{border-color:var(--accent)}
.option input:checked+label{border-color:var(--accent);background:#edf5eb;box-shadow:inset 0 0 0 1px var(--accent)}
body.dark .option input:checked+label{background:#203222}
.range-wrap{display:flex;align-items:center;gap:15px}
.range-wrap input{flex:1}
.range-value{min-width:76px;text-align:center;border:1px solid var(--line);border-radius:12px;padding:10px;background:var(--surface-solid);font-weight:700}
input[type=range]{accent-color:var(--accent)}
.select,input[type=number]{width:100%;padding:14px 15px;border:1px solid var(--line);border-radius:14px;background:var(--surface-solid);color:var(--text);outline:none}
.calc-actions{display:flex;justify-content:space-between;align-items:center;margin-top:17px}
.results-shell{padding-top:20px}
.results-header{display:flex;justify-content:space-between;align-items:flex-end;gap:20px;margin:24px 0 32px}
.results-header h2{font-size:54px;letter-spacing:-.05em;line-height:.98}
.result-hero{display:grid;grid-template-columns:1.1fr .9fr;gap:14px}
.score-card,.biggest-card{border:1px solid var(--line);border-radius:26px;background:var(--surface);padding:32px;min-height:240px}
.score-value{font-family:"Space Grotesk";font-size:79px;letter-spacing:-.07em;line-height:1;margin-top:15px}
.score-value small{font-family:"DM Sans";font-size:14px;letter-spacing:0;color:var(--muted);margin-left:10px}
.score-period{font-size:13px;color:var(--muted);margin-top:3px}
.score-note{margin-top:35px;color:var(--muted);font-size:12px}
.biggest-card{background:var(--dark);color:var(--bg);display:grid;grid-template-columns:1fr auto;align-content:center;column-gap:10px}
.biggest-card .mini-label{color:rgba(255,255,255,.55);grid-column:1/3}

```

```

.biggest-icon{font-size:32px;margin-top:15px}

.biggest-card h3{align-self:end;font-size:27px;margin-top:11px}

.biggest-number{font-family:"Space Grotesk";font-size:29px;align-self:end;margin-top:16px}

.biggest-card p{grid-column:1/3;color:rgba(255,255,255,.65);font-size:12px;line-height:1.5;margin:10px 0 0}

.breakdown-section,.recommendation-section,.whatif-section{margin-top:75px}

.breakdown-grid{display:grid;grid-template-columns:340px 1fr;gap:20px;margin-top:25px}

.chart-card,.bars-card{border:1px solid var(--line);background:var(--surface);border-radius:26px;padding:22px}

.chart-card{display:grid;place-items:center;position:relative;min-height:370px}

.chart-card canvas{max-width:310px;width:100%;height:auto}

.chart-center{position:absolute;display:flex;flex-direction:column;align-items:center;pointer-events:none}

.chart-center strong{font-family:"Space Grotesk";font-size:28px;letter-spacing:-.06em}

.chart-center span{font-size:10px;color:var(--muted)}

.bars-card{display:flex;flex-direction:column;justify-content:center;gap:22px}

.bar-row{display:grid;grid-template-columns:115px 1fr 65px;align-items:center;gap:10px}

.bar-label{font-size:12px;font-weight:700}

.bar-track{height:9px;background:var(--line);border-radius:99px;overflow:hidden}

.bar-fill{height:100%;border-radius:99px;background:var(--accent);transition:width .55s cubic-bezier(.2,.8,.2,1)}

.bar-number{font-size:12px;text-align:right;color:var(--muted)}

.rec-layout{display:grid;grid-template-columns:1.25fr .75fr;gap:15px;margin-top:25px}

.rec-main{border:1px solid var(--line);background:var(--surface);border-radius:26px;padding:30px}

.rec-main .rec-tag{font-size:11px;color:var(--accent);font-weight:800;letter-spacing:.12em}

.rec-main h3{font-size:28px;margin-top:10px}

.rec-main p{color:var(--muted);line-height:1.65}

.rec-list{display:grid;gap:9px;margin-top:22px}

.rec-item{display:flex;gap:12px;border-top:1px solid var(--line);padding-top:15px}

```

```

.rec-item span{width:25px;height:25px;border-
radius:8px;background:#e4eee3;display:grid;place-items:center;color:var(--accent);font-
weight:800;font-size:11px;flex:none}

body.dark .rec-item span{background:#293f2b}

.rec-item div strong{font-size:13px}.rec-item div p{font-size:12px;margin:2px 0 0}

.rec-side{display:grid;gap:15px}

.tiny-card{border:1px solid var(--line);background:var(--surface);border-
radius:22px;padding:23px}

.tiny-card>span{display:block;font-size:10px;color:var(--accent);font-weight:800;margin-
bottom:17px}

.tiny-card strong{font-size:18px;font-family:"Space Grotesk"}

.tiny-card p{font-size:12px;color:var(--muted);line-height:1.55}

.whatif-grid{display:grid;grid-template-columns:repeat(3,1fr);gap:12px;margin-top:25px}

.whatif-card{border:1px solid var(--line);background:var(--surface);border-
radius:20px;padding:20px;transition:.2s}

.whatif-card.selected{border-color:var(--accent);box-shadow:0 12px 30px
rgba(68,112,72,.12)}

.whatif-card button{width:100%;background:none;border:0;color:var(--text);text-
align:left;padding:0}

.whatif-card .whatif-icon{font-size:24px}.whatif-card h3{font-size:16px;margin:12px 0
6px}.whatif-card p{font-size:12px;color:var(--muted);line-height:1.5}

.whatif-saving{font-weight:800;color:var(--accent);font-size:12px}

.whatif-total{margin-top:13px;border-radius:18px;padding:17px 20px;background:var(--
dark);color:var(--bg);display:flex;justify-content:space-between;align-items:center;font-
size:12px}

.whatif-total strong{font-family:"Space Grotesk";font-size:22px}

.final-cta{margin-top:75px;padding:34px;border-radius:27px;background:var(--accent-
2);display:flex;justify-content:space-between;align-items:center;gap:25px}

.final-cta h2{font-size:31px;max-width:660px;margin-top:7px}

.final-cta p{color:var(--muted);max-width:600px;margin-bottom:0}

.site-footer{display:flex;justify-content:space-between;gap:20px;margin-top:38px;padding-
top:18px;border-top:1px solid var(--line);font-size:10px;color:var(--muted)}

@media(max-width:850px){

.hero{grid-template-columns:1fr;min-height:auto;padding:50px 0}.hero-
visual{height:360px}

```

```

.flow-grid {grid-template-columns: repeat(2, 1fr)}
.result-hero, .rec-layout {grid-template-columns: 1fr}
.breakdown-grid {grid-template-columns: 1fr}
.chart-card {min-height: 320px}
}

@media (max-width: 580px) {
  .site-header {height: 68px} .site-header nav a {display: none}
  .screen {padding-left: 15px; padding-right: 15px}
  h1 {font-size: 54px} .hero-text {font-size: 16px} .hero-actions {align-items: flex-start; flex-direction: column}
  .micro-stats {gap: 18px; flex-wrap: wrap}
  .hero-visual: before {width: 280px; height: 280px}
  .orbit-card {width: 250px; height: 250px} .orbit-number {font-size: 59px}
  .category-card {right: -6px; top: 35px} .tip-card {left: 0; bottom: 35px}
  .question-card {padding: 25px} .question-copy h2 {font-size: 34px}
  .option-grid, .whatif-grid {grid-template-columns: 1fr}
  .results-header {align-items: flex-start; flex-direction: column} .results-header h2 {font-size: 40px}
  .score-value {font-size: 60px} .bar-row {grid-template-columns: 90px 1fr 55px}
  .final-cta {flex-direction: column; align-items: flex-start}
  .site-footer {flex-direction: column}
}

```